

FLife

Operating Model Conditioning

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When conducting Management Strategy Evaluation the Operating Model (OM) describes the resource dynamics for use in simulation to provide a basis for the comparison of alternative assessment methods and management strategies. The Operating Model also includes the Observation Error Model (OEM), which generates simulated fishery data.

Operating Model Conditioning

Conditioning is the process of developing an simulation or Operating Model using the available data and knowledge. Operating Models should represent plausible alternative hypotheses, and reflect the concerns of stakeholders, since involving stakeholders is crucial to ensure that the Management Strategy Evaluation process contributes to managing fisheries in a way that respects both ecological and social aspects of sustainability Goethel et al. (2019)

A definition of a plausible scenario is one that fits prior knowledge, with many sources of corroboration, without the complexity of explanation, and with minimal conjecture (Connell and Keane 2006). Plausibility may be determined formally, e.g. based on a statistical approach to determine whether a system equivalent to the model generated the data (Laurence T. Kell et al. 2021) or on expert judgement (e.g., Leach et al. 2014).

Stock assessments are commonly used to condition OMs, however there is not always sufficient information to evaluate data quality and estimate model parameters (Board, Council, et al. 1998). A reason for this is because observations are seldom complete and variables of interest are latent. Therefore, there is often insufficient information to estimate key parameters Lee et al. (2011), and alternative choices may provide equally good fits (Barlow et al. 1972). For example, in an age-based assessment, the production function and hence reference points are largely determined by fixed parameters e.g. natural mortality, and growth and fecundity-at-age, and steepness Mangel et al. (2013).

The main sources of uncertainty in stock assessment are i) parameters for which there is little information in the data and so are fixed (e.g. natural mortality), ii) model structure, and iii) future process variability, e.g. recruitment, (A. E. Punt et al. 2016). Therefore to ensure robustness often requires conducting sensitivity tests or building ensembles based on alternative assessment scenarios, for example using hypotheses based on life history theory (Hamel et al. 2023). Life history theory has also been used tpo generate OMs for data limited stocks see Fischer, De Oliveira, and Kell (2020), Walker et al. (2023).

If alternative models explain the data equally well, it does not necessarily follow that a model provides a true account of how the data arose (Hodges, Dewar, and Center 1992) Therefore to evaluate robustness, management strategies must be tested under a range of hypothetical but plausible scenarios. For example, does a harvest strategy only achieve management objectives if natural mortality and the form of the stock recruitment relationship are assumed to be known.

In stock assessment sensitivity and uncertainty analyses are often used to understand the impacts of decisions in the face of uncertainty. In sensitivity analysis, a limited set of scenarios, which include the most important uncertainties, are considered, while in uncertainty analysis, the objective is to quantify the probability of specific output values. In both cases, procedures adopted for the selection, rejection and weighting of scenarios

will affect the robustness of advice. Therefore, a cherry-picking approach for choosing model configurations based on the implications for management advice should be avoided. For this reason how the choice of OMs is important as this will determine the performance of a management strategy.

Therefore when conditioning OMs a reference set and robustness sets are normally agreed in advance, before running the simulation trials.

Reference Set

When conditioning Operating Models, it is common practice to condition a reference set, i.e. a limited set of scenarios, which include the most important uncertainties in the model structure, parameters, and data. These comprise alternative scenarios which have both high plausibility, reflect the concerns of stakeholders, and are likely to major impacts on the performance of Management Strategies.

Instead of conditioning a number of OMs for the reference set, an alternative is to implement a single scenario but with priors that reflect uncertainty in parameters. Only one OMs is implemented because multiple simulation runs are not comparable with the estimation of posterior distributions in a Bayesian analysis Berger et al. (1994). A potential problem with such an approach is lack of considering variations in process and measurement errors among simulation runs.

Robustness Set

A robustness set examines performance beyond the range of the Reference Set. These are typically considered to be less likely than the reference set Operating Models, and often focus on particularly challenging circumstances with potentially negative consequences to be avoided. A benefit of the reference set is that the value-of-information can be evaluated, i.e. is there a benefit in reducing uncertainty by conducting scientific study or collecting improved data.

Redundancy

MSE is a form of classification problem, used to identify the best performing management strategy based on a set of performance metrics or statistics (Fischer et al. 2021). When there are multiple OMs and metrics these may overlap, resulting in imbalances, which may mean result in inefficiency or bias. Maillo, Triguero, and Herrera (2020) have provided a package of metrics for big data classification problems. In particular, Neighborhood Density and Decision Tree Progression, which study density and accuracy progression. In addition, they have proposed a number of basic metrics that can quickly characterise datasets.

Methods

Operating Models are generally developed using a stock assessment paradigm. However, there are many processes for which there is little information for in stock assessment datasets, e.g. natural mortality, the steepness of the stock-recruitment relationship, the form of population regulation and density dependence. Therefore, life history relationships were used to create Operating Models based on life-history relationships. This allows operating models to be conditioned on ecological processes, of particular value in data-poor situations where knowledge and data are limited, but also in data-rich situations as simulation testing an assessment procedure using a model conditioned on the same assumptions is not necessarily a true test of robustness.

To create an Operating Model, the FLR (L. T. Kell et al. 2007) packages mydas and FLife were used. First an equilibrium per-recruit model was parameterised for growth, maturity and natural mortality-at-age. The per-recruit model then was combined with a stock recruit relationship (Beverton and Holt 1993).

A reference set was conditioned using a multivariate distribution for $x, y, \dots$ to model uncertainty about parameters. While a reference set was conditioned to represent uncertainty about the form of processes. Virgin biomass was set at a constant value across all stocks and scenarios, so that results are comparable, i.e. presented in terms of exploitation level and relative stock size and catch.

Life history parameters for growth, natural mortality and maturity were used to develop an age-based Operating Model by parameterising functional forms for weight (W), proportion mature (Q), natural mortality (M) and fishing mortality (F) at age. These were then used to calculate spawner (S/R) and yield-per-recruit (Y/R), which when combined with a stock-recruitment relationship (Sissenwine and Shepherd 1987) can be used to calculate the equilibrium stock size as a function of fishing mortality (F). The equilibrium relationships can then be turned into a dynamic model and projected forward.

An equilibrium per-recruit model was parameterised for growth, maturity and natural mortality-at-age. The per-recruit model was combined with a stock-recruit relationship (Beverton and Holt 1993). An age-structured simulation model was conditioned using life history theory to provide a theoretical basis for developing hypotheses about population dynamics. The parameters were: growth model parameters (k , L_∞ , t_0 , Von Bertalanffy 1957); a and b of the length-weight relationship; and the length at which 50% were mature (L_{50}). Natural mortality-at-age was modelled as a function of length (Gislason et al. 2008) and spawning stock biomass (SSB) was used as a proxy for stock reproductive potential (SRP, Trippel 1999). It was assumed that fecundity is proportional to the weight-at-age of the sexually mature portion of the population irrespective of the demographic composition of adults (Murawski, Rago, and Trippel 2001) and that processes such as sexual maturity are simple functions of age and independent of sex (Matsuda et al. 1996).

A variety of functional forms can be assumed for growth, natural mortality, maturity, the selection pattern of the fisheries and the stock-recruitment relationship. Processes such as growth and maturity-at-age are generally observable, and data can be collected to identify the correct functional form and to estimate parameters. The functional form and parameters for processes such as natural mortality and the stock-recruitment relationship are typically poorly known (Michielsens and McAllister 2004). Therefore, assumptions have to be made and to evaluate the sensitivity of any analysis to those assumptions a variety of scenarios need to be considered.

Historical exploitation was simulated where

Scenarios

Even for data-rich stock assessments there is often large uncertainty about the dynamics i.e. model uncertainty, (A. Punt 2008), for example estimates of stock status are highly sensitive to assumptions about natural mortality-at-age (Jiao et al. 2012), vulnerability of age classes to the fisheries (Brooks, Powers, and Cortés 2009), and the relationship between stock and recruitment (Cury et al. 2014). Therefore, a reference case was defined and a robustness set developed representing the main sources of uncertainty (Table ??);

Selectivity depends on the vulnerability of individuals to fishing, and is typically either asymptotic or dome-shaped. The former indicates an initial increase with age or size followed by a levelling off, while in the later case selectivity-at-age declines. Selectivity will differ between fisheries depending on gear characteristics, when and where the fishery operates and the biology of the species. For example, estimates for species captured with gill nets or hooks will be biased if the model incorrectly assumes logistic selectivity because the missing large fish will be assumed to have been caught, whereas they could just be missed by the gear. Fisheries selectivity was therefore modelled as a double normal, as this allowed both asymptotic and dome shaped selectivity to be simulated. Logistic selectivity in the base case was based on the maturity ogive, so that MSY reference points are comparable across case studies. The consequences of shifting the selection patterns is well known, since if you fish below or above L_{opt} (the length at which a cohort attains its maximum biomass) you reduce MSY , while if you take fish before L_{50} F_{MSY} is reduced and B_{MSY} is increased as older fish need to be conserved. Therefore, scenarios were not case-specific

Life history processes

Individual Growth Growth in length in this example was assumed to follow the Von Bertalanffy growth equation Von Bertalanffy (1957)

$$L = L_\infty(1 - \exp(-k(t - t_0))) \quad (1)$$

where k is the rate at which the rate of growth in length declines as length approaches the asymptotic length L_∞ and t_0 is the hypothetical time at which an individual is of zero length.

Length is converted to mass using the length-weight relationship

$$W = aL_t^b \quad (2)$$

where a is the condition factor and b is the allometric growth coefficient.

Maturity-at-age Proportion mature-at-age is modelled by the logistic equation with 3 parameters: age at 50% (a_{50}), 95% (a_{95}) mature, and maturity as age approaches infinity.

$$f(x) = \begin{cases} 0 & \text{if } (a_{50} - x)/a_{95} > 5 \\ a_\infty & \text{if } (a_{50} - x)/a_{95} < -5 \\ \frac{m_\infty}{1.0 + 19.0^{(a_{50} - x)/95}} & \text{otherwise} \end{cases} \quad (3)$$

Natural Mortality The natural mortality of exploited fish populations is often assumed to be a species-specific constant independent of body size. This assumption has important implications for size-based fish population models and for predicting the outcome of size-dependent fisheries management measures such as mesh-size regulations (Gislason et al. 2008). Direct estimates of the instantaneous natural mortality made in controlled studies, however, are shown to vary by age (Lorenzen and Enberg 2002). Although M can sometimes be estimated within an assessment model, for example where data from tagging provide information independent of fishing mortality rates. However, in most cases, M is derived from life-history relationships, e.g. based on size Griffiths and Harrod (2007). The large and ever-increasing literature on this subject is a reflection of uncertainty. Gislason et al. (2008) in an empirical study showed that M is significantly related to body length, asymptotic length and k . Temperature is non-significant when k is included, since k itself is correlated with temperature, i.e.

$$M = 0.55L^{1.61}L_\infty^{1.44}k \quad (4)$$

Selection Pattern In this study, the vulnerability of individuals to fishing was modelled by the selection pattern. Typically, selectivity is either asymptotic or dome-shaped. The former indicates an initial increase with age or size followed by a levelling off, while in the latter case selectivity-at-age declines. Selectivity will differ between fisheries depending on gear characteristics, when and where the fishery operates and the biology of the species. Fisheries selectivity was therefore modelled as a double normal, as this allowed both asymptotic and dome-shaped selectivity to be simulated. The double normal has three parameters that describe the age at maximum selection (a_1), the rate at which the left-hand limb increases (sl) and the right-hand limb decreases (sr) which allows flat topped or domed shaped selection patterns to be chosen, i.e.

$$f(x) = \begin{cases} 2^{-\left(\frac{a-a_1}{sl}\right)^2}, & \text{if } a < a_1 \\ 2^{-\left(\frac{a-a_1}{sr}\right)^2}, & \text{if } a \geq a_1 \end{cases} \quad (5)$$

Spawning Stock Biomass Spawning stock biomass (S), is the sum of the products of the numbers of females, N , proportion mature-at-age, Q and their mean fecundity-at-age, G , which is taken to be proportional to their weight-at-age i.e.

$$S = \sum_{i=0}^p N_i Q_i G_i \quad (6)$$

In this study, fecundity-at-age is assumed proportional to biomass and the sex ratio to be 1:1.

Stock Recruitment Relationship A Beverton and Holt stock-recruitment relationship Beverton and Holt (1993) was assumed. This relationship is derived from a simple density-dependent mortality model where the more survivors there are the higher the mortality. It is assumed that the number of recruits (R) increases towards an asymptotic level (R_{max}) as S increases, i.e.

$$R = Sa/(b + S) \quad (7)$$

The relationship was reformulated in terms of steepness (h), virgin biomass (v) and $S/R_{F=0}$. Where steepness is the proportion of the expected recruitment produced at 20% of virgin biomass relative to virgin recruitment (R_0). However, there is often insufficient information to allow its estimation from stock assessment (**pepin2015reconsidering?**) and so scenarios were considered for different value of steepness, i.e. resilience to overfishing. Virgin biomass was set at 1000 Mt to allow comparisons to be made across scenarios.

$$R = \frac{0.8R_0h}{0.2S/R_{F=0}R_0(1-h) + (h-0.2)S} \quad (8)$$

Equilibrium Analysis

Sissenwine and Shepherd (1987), estimated surplus production using an age-based analysis using an equilibrium analysis by combining a stock-recruitment relationship, a spawning-stock-biomass-per-recruit analysis, and a yield-per-recruit analysis. For any specified rate of fishing mortality, an associated value of spawning stock biomass (S) per recruit (R) is S/R is defined, based on the assumed processes for growth, natural mortality and selection pattern-at-age detailed in the previous sections.

$$S/R = \sum_{i=0}^{p-1} \frac{\sum_{j=0}^{i-1} -F_j - M_j}{e^{j=0}} W_i Q_i + e^{\sum_{i=0}^{p-1} -F_i - M_i} \frac{W_p Q_p}{1 - e^{-F_p - M_p}} \quad (9)$$

When the value of S/R obtained is inverted and superimposed on the stock-recruitment function as a slope (R/S), the intersection of this slope with the stock-recruitment function defines an equilibrium level of recruitment. When this value of recruitment is multiplied by the yield per recruit calculated for the same fishing mortality rate, the equilibrium yield associated with the fishing mortality rate emerges (**gabriel1999review?**).

$$Y/R = \sum_{a=r}^{n-1} e^{\sum_{i=r}^{a-1} -F_i - M_i} W_a \frac{F_a}{F_a + M_a} (1 - e^{-F_i - M_i}) + e^{\sum_{i=r}^{n-1} -F_i - M_i} W_n \frac{F_n}{F_n + M_n} \quad (10)$$

The second term is the plus-group, i.e. the summation of all ages from the last age to infinity.

Projection

The stock-recruitment relationship and the vectors of weight, natural mortality, maturity and selectivity-at-age allow a forward projection model to be created, which forms the basis of the Operating Model.

$$N_{t,a} = \begin{cases} R_t, & \text{if } a = 0, \\ N_{t-1,a-1}e^{-Z_{t-1,a-1}}, & \text{if } 1 \leq a \leq A-1, \\ N_{t-1,A-1}e^{-Z_{t-1,A-1}} + N_{t-1,A}e^{-Z_{t-1,A}}, & \text{if } a = A, \end{cases} \quad (11)$$

Where $N_{t,a}$ is the number of fish of age a at the beginning of year t , R_t is the total number of recruits born in year t . Here, A is the so-called plus group age, which is an aggregated age greater than or equal to the actual age A .

Seasonal Operating Model

A seasonal Operating Model was developed to explore the impact of shorter lags between survey, advice and exploitation. This allowed alternative management options such as in-season rules to be considered.

Stock Components or Mixed Stocks

Software Versions

- R version 4.4.2 (2024-10-31 ucrt)
- FLCore: 2.6.20.9333
- FLPKG:
- FLBRP: 2.5.9.9051
- FLasher: 0.7.1.9223
- ggplotFL: 2.7.0.9139
- FLife: 3.4.0
- mydas: 1.2.2
- FLCandy: 1.6.0
- **Compiled:** Mon Apr 7 18:12:03 2025
- **Git Hash:** '95cd199'

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